

→ MSE : for normal random variable

$$MSE_{\hat{\theta}} = \text{Var}(x) + \text{Bias}^2$$

$$\text{Var}(x) = [E(\hat{\theta}^2)] - [E(\hat{\theta})]^2$$

$$\text{Bias} = [E(\hat{\theta}) - \theta]^2$$

→ MSE for different distributions:

Bernoulli:

$$\text{Bias} = 0, E(x) = p = E(\hat{p})$$

$$MSE(\hat{p}) = \text{Var}(\hat{p}) = p(1-p)$$

$p \rightarrow$ probability

Binomial: $E[\hat{p}] = E\left[\frac{x}{n}\right] = \frac{np}{n} = p$

$$MSE(\hat{\theta}) = \frac{p(1-p)}{n}; \text{Bias} = 0$$

Poisson:

$$\text{Bias} = 0$$
$$MSE(\hat{\mu}) = \frac{\mu}{n}$$

Normal:

$$E(\hat{\mu}) = \mu \quad [\text{bias} = 0]$$
$$\text{Var}(\hat{\mu}) = \frac{\sigma^2}{n}$$

$$\Rightarrow MSE(\hat{\mu}) = \frac{\sigma^2}{n}$$

→ Likelihood function:

→ Bernoulli:

PMF $P(X=x) = p^x(1-p)^{1-x}$

MLE of p is:

$$\hat{p} = \bar{x}$$

→ Binomial:

$$P = \frac{x}{n}$$

MLE of p is $\hat{p} = \frac{x}{n}$

MLE of $\frac{p}{1-p} = \frac{\hat{p}}{1-\hat{p}} = \frac{x}{n-x}$

→ Poisson:

MLE of λ is $\hat{\lambda} = \bar{x}$

→ Normal:

MLE of μ is $\hat{\mu} = \bar{x}$

MLE of σ is:

$$\hat{\sigma} = \sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}$$

Square σ for Variance.

Confidence interval

→ two-sided confidence interval :

$$\text{Sample statistic} \pm [\text{Critical value} * \text{Standard error}]$$

→ one-sided :

Lower limit :

$$\text{Sample mean} - (\text{Critical value} * \text{Standard error})$$

Upper limit :

$$\text{Sample mean} + [\text{Critical value} * \text{Standard error}]$$

→ Two sided :

$$\bar{x} \pm Z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}$$

→ one sided :

Upper bound / Lower Interval :

$$(-\infty, \bar{x} + Z_{\alpha} \cdot \frac{s}{\sqrt{n}})$$

Lower bound / Upper Interval

$$(\bar{x} - Z_{\alpha} \cdot \frac{s}{\sqrt{n}}, \infty)$$

→ In all cases 's' can be replaced with 'σ' if known.

These formula to be applied when sample size is large [$n \geq 30$]

→ Width of a confidence interval :

$$\begin{aligned}\text{width} &= \text{upper limit} - \text{lower limit} \\ &= 2 \times Z_{\alpha/2} \frac{S}{\sqrt{n}}\end{aligned}$$

→ Traditional method for computing confidence interval for a proportion :

$$\hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$\hat{p} = \frac{\text{No. of Successes}}{n}$$

The method cannot be used unless the sample contains at least 10 successes and 10 failures.

→ Agresti-Coull method for proportion :
[for any sample size 'n']

$$\tilde{p} \pm Z_{\alpha/2} \cdot \sqrt{\frac{\tilde{p}(1-\tilde{p})}{\tilde{n}}}$$

$$\tilde{n} = n + 4$$

$$\tilde{p} = \frac{x + 2}{\tilde{n}}$$

→ Large-Sample point Estimators [to Estimate parameter] [one-sided interval]

To estimate one of four population parameters when the sample sizes are large, use the following point estimators with the appropriate margins of error.

Parameter	Point Estimator	Margin of Error
μ	$\bar{x}$	$\pm 1.96 \left(\frac{s}{\sqrt{n}} \right)$
p	$\hat{p} = \frac{x}{n}$	$\pm 1.96 \sqrt{\frac{\hat{p}\hat{q}}{n}}$
$\mu_1 - \mu_2$	$\bar{x}_1 - \bar{x}_2$	$\pm 1.96 \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$
$p_1 - p_2$	$(\hat{p}_1 - \hat{p}_2) = \left(\frac{x_1}{n_1} - \frac{x_2}{n_2} \right)$	$\pm 1.96 \sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}$

→ large - sample point Estimator [for population parameters] [Two sided / bound interval] :

To estimate one of four population parameters when the sample sizes are large, use the following interval estimators.

Parameter	$(1 - \alpha)100\%$ Confidence Interval
μ	$\bar{x} \pm z_{\alpha/2} \left(\frac{s}{\sqrt{n}} \right)$
p	$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}}$
$\mu_1 - \mu_2$	$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$
$p_1 - p_2$	$(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}$

→ Estimating the difference between two means

$$\bar{x} - \bar{y} \pm z_{\alpha/2} \sqrt{\frac{\sigma_x^2}{n_x} + \frac{\sigma_y^2}{n_y}}$$

σ_x can be replaced with s_x if σ is unknown.
 $[n_x \& n_y \geq 30]$

→ Students t - distribution:

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

with " $n-1$ " degrees of freedom denoted by " t_{n-1} ".

→ Note: This is only used when sample data is small. If large sample data is used, the ' t ' curve approaches normal curve, so we just use Normal distribution.

' n ' is large if $[n \geq 30]$

→ Note: It only gives values for " $t > n$ " for " $t < n$ " find $t > n$ and make it negative.

→ Linear interpolation for ' t '

$$y_1 = \frac{x - x_1}{x - x_2} (y_1 - y_2)$$

→ Confidence interval for a small sample:

$$\bar{x} \pm t_{n-1, \alpha/2} \frac{s}{\sqrt{n}}$$

$$\alpha = [100\% - \text{Confidence-level}\%]$$

→ Confidence interval between two proportions

Agresti - Caffo method:

$$p_x - p_y = \tilde{p}_x - \tilde{p}_y \pm Z_{\alpha/2} \sqrt{\frac{\tilde{p}_x(1-\tilde{p}_x)}{\tilde{n}_x} + \frac{\tilde{p}_y(1-\tilde{p}_y)}{\tilde{n}_y}}$$

$$\tilde{p}_x = \frac{(x+1)}{\tilde{n}_x} \quad \tilde{p}_y = \frac{(y+1)}{\tilde{n}_y}$$

$$\tilde{n}_x = n_x + 2 \quad \tilde{n}_y = n_y + 2$$

→ Confidence interval for true mean difference

• If n is small [$n \leq 30$]

$$\bar{D} \pm t_{n-1, \alpha/2} \frac{s_D}{\sqrt{n}}$$

• If n is large [$n > 30$]

$$\bar{D} \pm Z_{\alpha/2} \frac{s_D}{\sqrt{n}}$$